MTH 451/551
Lecture 10

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1 Numerical Integration and Differentiation: Basic Quadrature

[Textbook reference: §4.1]

We previously saw the midpoint rule for approximating integrations,

$$\int_a^b f(x)\,dx = (b-a)f\left(\frac{a+b}{2}\right) + \mathcal{O}((b-a)^3),$$

for $f \in C^2[a,b]$. This is one example of **numerical quadrature**, which is a term that refers to the approximation of integrals through computable finite sums. We have focused on *polynomial approximation* for good reason. Most of the quadratures we will considered will be based on the idea of approximation f first by a polynomial p, and then integrating p exactly, so that $\int f \approx \int p$.

Definition 1 (Quadrature; Textbook Definition 4.1)**.** *For $f \in C[a,b]$, a **quadrature approximation** of $I(f) = \int_a^b f(x)\,dx$ is a linear combination of values of f at distinct points $\{x_0, x_1, \dots, x_n\} \subset [a,b]$,*

$$I(f) \approx Q(f) = \sum_{j=0}^{n} w_j f(x_j).$$

*We refer to the nodes x_j as **quadrature points**, and the numbers w_j as **quadrature weights**. We say that a quadrature Q is **exact on** $\mathbb{P}_m$ when $Q(p) = I(p)$ for all $p \in \mathbb{P}_m$.*

Lemma 1 (Textbook Lemma 4.2)**.** *Both I and Q are linear functionals on $C[a,b]$. In other words, for all $f, g \in C[a,b]$ and $\alpha, \beta \in \mathbb{R}$,*

$$I(\alpha f + \beta g) = \alpha I(f) + \beta I(g) \quad \text{and} \quad Q(\alpha f + \beta g) = \alpha Q(f) + \beta Q(g).$$

Consequently, Q is exact on $\mathbb{P}_m$ if and only if $Q(p_j) = I(p_j)$ for every element of a basis $\{p_0, \dots, p_m\}$ of $\mathbb{P}_m$.

Definition 2 (Interpolatory and Newton–Cotes Quadrature; Textbook Definition 4.3)**.** *Given $f \in C[a,b]$ and distinct quadrature points $\{x_0, x_1, \dots, x_n\} \subset [a,b]$, the corresponding* ***interpolatory quadrature*** *is obtained by integrating the Lagrange interpolating polynomial $p \in \mathbb{P}_n$ exactly:*

$$Q(f) = \int_a^b p(x)\,dx = \sum_{j=0}^{n} w_j f(x_j), \qquad \text{where} \qquad w_j = \int_a^b \ell_j(x)\,dx,$$

where $\{\ell_j\}$ are the Lagrange polynomials for the quadrature points. By construction, an interpolatory quadrature is exact on $\mathbb{P}_n$. In the special case that the quadrature points are evenly spaced,

$$x_j = a + \frac{j}{n}(b-a), \qquad 0 \le j \le n,$$

the rule is called a ***(closed) Newton–Cotes quadrature****.*

Lemma 2 (Textbook Definition 4.3)**.** *Any $(n+1)$-point quadrature that is exact on $\mathbb{P}_m$ for some $m \ge n$ must be the interpolatory quadrature for its quadrature points.*

Proof. Let $\{x_i\}$ denote the quadrature points. Since Q is exact, $Q(\ell_j) = \int_a^b \ell_j(x)\,dx$. By the definition of Q,

$$Q(\ell_j) = \sum_{k=0}^{n} w_k \ell_j(x_k) = \sum_{k=0}^{n} w_k \delta_{jk} = w_j,$$

and so $w_j = \int_a^b \ell_j(x)\,dx$, implying that Q is interpolatory. □

Theorem 1 (Textbook Theorem 4.4)**.** *Suppose that an $(n+1)$-point quadrature Q is exact on $\mathbb{P}_m$ for some $m \ge n$. If $q \in \mathbb{P}_m$ satisfies $q(x_j) = f(x_j)$ for $0 \le j \le n$, then*

$$I(f) - Q(f) = I(f - q).$$

Proof. By the previous lemma, if a quadrature Q of the form (4.1) is exact on $\mathbb{P}_m$ for some $m \ge n$, then it is necessarily an interpolatory quadrature (4.2). Since q interpolates f at the quadrature points, and Q is exact on $\mathbb{P}_m$, we have $I(f) - Q(f) = I(f) - Q(q) = I(f) - I(q) = I(f - q)$. □

The above theorem implies that the *quadrature error* is given by the *integral of the interpolation error*. Therefore, our previous results bounding the interpolation error can be used to derive error estimates for quadratures.

Theorem 2 (Five Basic Quadrature Rules; Textbook Theorem 4.5)**.** *Given a bounded interval $[a,b]$, let $c = (a+b)/2$. Five basic quadrature rules, $Q(f) \approx I(f) = \int_a^b f(x)\,dx$, together with their associated errors, are given in the table below (Textbook Table 4.1). For each error formula, $\xi \in (a,b)$. The regularity assumption on f required for each error formula is given in the final column of the table.*

Name	Quadrature $Q(f)$	Error $I(f) - Q(f)$	Regularity
Left Endpoint	$(b-a)f(a)$	$\frac{f'(\xi)}{2}(b-a)^2$	$f \in C^1[a,b]$
Right Endpoint	$(b-a)f(b)$	$-\frac{f'(\xi)}{2}(b-a)^2$	$f \in C^1[a,b]$
Trapezoid	$\frac{b-a}{2}\big(f(a)+f(b)\big)$	$-\frac{f''(\xi)}{12}(b-a)^3$	$f \in C^2[a,b]$
Midpoint	$(b-a)f(c)$	$\frac{f''(\xi)}{24}(b-a)^3$	$f \in C^2[a,b]$
Simpson	$\frac{b-a}{6}\big(f(a)+4f(c)+f(b)\big)$	$-\frac{f^{(4)}(\xi)}{90}\left(\frac{b-a}{2}\right)^5$	$f \in C^4[a,b]$